



ASP.NET CORE CHEAT SHEET

Core setup, initialization, operations & integration quick reference

PART 1: SETUP & CORE

01. SETUP & PROGRAM.CS

Installation

```
var builder = WebApplication.CreateBuilder(ar...
builder.Services.AddControllers();
builder.Services.AddEndpointsApiExplorer();
builder.Services.AddSwaggerGen();
var app = builder.Build();
app.UseRouting(); app.MapControllers();
app.Run();
```

04. ROUTING

Workflow

```
[Route("api/[controller]/[action]")]
[HttpGet("{id:int}")]
[HttpGet("{slug:regex^[a-z-]+$}")]
app.MapControllerRoute("default", "{controlle...
app.MapGet("/", () => "Hello World!");
```

02. MINIMAL APIS

Architecture

```
app.MapGet('/users', async (UserRepo repo) =>
await repo.GetAllAsync());
app.MapPost('/users', async (User u, UserRepo...
await r.AddAsync(u);
return Results.Created($"{u.Id}", u);
});
app.MapPut('/users/{id}', ...).RequireAuthori...
```

05. DEPENDENCY INJECTION

CRUD API

```
builder.Services.AddScoped<UserService, User...
builder.Services.AddSingleton<ICache, MemoryC...
builder.Services.AddTransient<IEmailSender, S...
public class Ctrl(IUserService svc) { ... } /...
```

Lifetimes: transient(new each time), Scoped(per request), Singleton(app lifetime)

03. CONTROLLERS

YAML / Config

```
[ApiController] [Route("api/[controller]")]
public class UsersController : ControllerBase...
[HttpGet] public async Task<IActionResult> Ge...
=> Ok(await _svc.GetAllAsync());
[HttpPost] public async Task<IActionResult> C...
=> CreatedAtAction(nameof(GetById), new{id=....
}
```

06. MIDDLEWARE

Integration

```
app.Use(async (ctx, next) => {
// before
await next(ctx);
// after
});
app.UseExceptionHandler('/error');
app.UseHsts(); app.UseHttpsRedirection();
app.UseAuthentication(); app.UseAuthorization...
```

Order matters: HTTPS Routing Auth Endpoints



SECURITY, MONITORING & OPERATIONS

Production hardening, observability, performance tuning & CLI reference

PART 2: OPS & SECURITY

07. CONFIGURATION

Hardening

```
var conn = builder.Configuration.GetConnectio...
var secret = builder.Configuration["Jwt:Secre...
builder.Services.Configure<AppSettings>(...
builder.Configuration.GetSection("App"));
// appsettings.json overridden by appsettings...
// Environment variables override config files
```

10. FILTERS

Unit Tests

```
[ServiceFilter(typeof(LoggingFilter))]
public class LoggingFilter : IActionFilter {
    public void OnActionExecuting(ActionExecuting...
    public void OnActionExecuted(ActionExecutedCo...
}
```

Filter types: Authorization, Resource, Action, Exception, Result

08. MODEL BINDING

Observability

```
[FromRoute] int id
[FromQuery] string? filter
[FromBody] CreateUserDto dto
[FromHeader(Name="X-API-Key")] string apiKey
[FromForm] IFormFile file
[FromServices] IService svc // DI in action ...
```

11. LOGGING

Commands

```
private readonly ILogger<UsersController> _lo...
_logger.LogInformation("Getting user {Id}", i...
_logger.LogError(ex, "Failed to create user");
// appsettings.json:
"Logging": { "LogLevel": { "Default": "Inform...
builder.Logging.AddConsole().AddDebug();
```

09. AUTH & AUTHORIZATION

Optimization

```
builder.Services.AddAuthentication(JwtBearerD...
.AddJwtBearer(opts => { opts.TokenValidationP...
[Authorize] [Authorize(Roles="Admin")]
[AllowAnonymous]
builder.Services.AddAuthorization(opts =>
opts.AddPolicy("Admin", p => p.RequireRole("A...)
```

12. COMMON GOTCHAS

Warning

UseAuthentication() must come before UseAuthorization()
AddDbContext() is Scoped don't inject into Singleton
async void in controllers hides exceptions
Nullable annotations on DTOs prevent null surprises
CORS must be configured before UseRouting()